

- 7 (a) Explain what is meant by *work function* of a metal by reference to the photoelectric effect.

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..... [1]

- (b) In a photoelectric emission experiment, light of wavelength 540 nm and intensity 150 Wm^{-2} is incident on a metal surface of work function 1.9 eV in an evacuated tube, such that an area of 14 mm^2 is illuminated.

- (i) Calculate the energy of the incident photon.

energy = J [2]

- (ii) Calculate the rate of photons incident on the metal surface.

rate = s^{-1} [2]

- (iii) Determine the maximum kinetic energy of the electrons emitted from the surface.

maximum kinetic energy = J [2]

- (c) An X-ray spectrum produced by copper is shown in Fig. 7.1 where I is the intensity and λ is the wavelength.

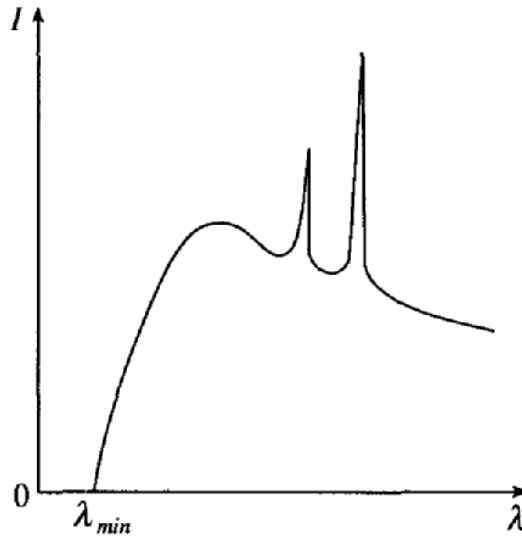


Fig. 7.1.

- (i) Explain how a continuous spectrum is formed and why is there a minimum wavelength.

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 [3]

- (ii) Sketch on Fig. 7.1 the new X-ray spectrum when a larger accelerating voltage is used instead. Explain your answer.

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 [3]